Arrhythmia Detection Using Machine Learning Algorithms With SMOTE Technique

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***Abstract*—Cardiac arrhythmia is an irregular heartbeat that can lead to severe cardiovascular conditions if undetected. Electrocardiogram (ECG) signals are widely used for arrhythmia classification, but manual diagnosis is time-consuming and subjective. In this study, we propose an automated arrhythmia detection system using Logistic Regression, Random Forest, LGBM, and XGBoost, combined with Synthetic Minority Over-sampling Technique (SMOTE) and Without SMOTE to address class imbalance in ECG data. We use the MIT-BIH Arrhythmia Database and evaluate model performance using accuracy, precision, recall, F1-score, and Kappa evaluation. Our findings indicate that the use of ensemble-based machine learning models, combined with appropriate data balancing techniques, enables accurate ECG-based arrhythmia classification, achieving an overall accuracy of 99.84%.**

***Index Terms*—Electrocardiogram (ECG), arrhythmia classification, ensemble machine learning, class imbalance handling, SMOTE, MIT-BIH Arrhythmia Database.**

I. Introduction

Cardiovascular diseases remain one of the leading causes of mortality worldwide, contributing to millions of deaths every year [1]. Early identification of abnormalities in heart rhythm is therefore essential for reducing medical risks. The ECG is widely used as a standard diagnostic tool because it provides a detailed recording of the heart's electrical activity in a non-invasive manner [2]. However, manual interpretation of ECG signals can be time-consuming, requires trained specialists, and becomes increasingly challenging during continuous long-term monitoring [3].

ECG recordings often contain various forms of noise, including baseline drift, muscle artifacts, and power-line interference, all of which can distort important waveform features and hinder accurate diagnosis [1]. Another major issue is class imbalance in ECG datasets. In databases such as MIT-BIH, normal beats significantly outnumber abnormal ones, causing machine learning models to prioritize majority classes and miss clinically significant arrhythmias [3]. This imbalance often leads to poor recall and low sensitivity for minority arrhythmia types, which limits the reliability of automated classification systems [4]. Additionally, natural variations in ECG morphology among different patients make consistent feature extraction even more challenging.

Machine learning approaches have gained significant attention for their potential to automate ECG interpretation [5] . Traditional algorithms such as Logistic Regression and Random Forest have shown promising results when supported by meaningful feature extraction techniques [6]. More recent boosting-based models, including LightGBM and XGBoost, have demonstrated strong classification performance and high computational efficiency in biomedical applications [7], [8]. To address class imbalance, the SMOTE has become a widely adopted approach for generating artificial minority samples and improving the representation of rare arrhythmias [9].

II. Literature Survey

The MIT-BIH Arrhythmia Database has been the primary benchmark for evaluating automated arrhythmia detection systems due to its well-annotated recordings and diverse heartbeat categories [2]. PhysioNet has further supported ECG research by providing standardized datasets and essential signal-processing tools that enable consistent experimentation across studies [10].

Early machine learning approaches relied on handcrafted ECG features. De Chazal et al. showed that morphological and interval-based descriptors—such as QRS duration and RR intervals—can effectively support heartbeat classification using traditional classifiers [11]. Similar feature-driven studies demonstrated that models such as SVM and statistical learners perform reasonably well when meaningful ECG attributes are extracted from preprocessed signals [12].

With the progression of deep learning, several studies shifted toward architectures capable of learning features directly from raw ECG data [13] . Acharya et al. introduced one of the early convolutional neural networks (CNN)based models for heartbeat classification, demonstrating that deep networks can automatically extract relevant morphological patterns without manual feature design [6]. Subsequent work enhanced these methods using wavelet transforms and recurrent models to capture temporal changes in ECG traces [14]. Patient-specific CNN frameworks have also been explored to improve adaptability and real-time detection performance [15].

Parallel to deep learning, ensemble learning methods—particularly tree-based models—have gained popularity for their efficiency and strong predictive performance on structured ECG features. LightGBM demonstrated high accuracy and speed in ECG classification tasks when combined with optimized feature sets [7], while XGBoost consistently outperformed several traditional models due to its robust boosting framework [8]. Random Forest also remains widely used because of its ability to handle heterogeneous features and its interpretability compared to deep learning models [16].

A major challenge highlighted in previous research is the severe class imbalance present in ECG datasets, where normal beats significantly outnumber abnormal ones [2]. This imbalance reduces the sensitivity of machine learning models to minority arrhythmias, leading to poor recall and misclassification of clinically important heartbeat types. To address this issue, SMOTE has been widely applied to oversample minority classes by generating synthetic samples that improve class representation during training [9]. Studies incorporating SMOTE reported noticeable improvements in detecting rare arrhythmias and overall F1-scores for minority classes [4], [17].

Overall, the existing literature demonstrates substantial advancements in machine learning-based arrhythmia detection but also indicates persistent challenges such as noise sensitivity, inter-patient variability, and the class imbalance problem. These limitations motivate continued research on improved preprocessing methods, balanced training strategies, and comparative assessment of machine learning models for reliable ECG classification.

The general workflow of ECG-based arrhythmia detection typically involves multiple stages, starting from signal acquisition and preprocessing, followed by heartbeat segmentation and feature extraction. Machine learning models are then trained to classify normal and abnormal heartbeats, often addressing class imbalance using oversampling techniques such as SMOTE. Finally, model performance is evaluated using standard classification metrics to assess accuracy and reliability. Based on this general workflow, recent studies have focused on improving classification performance through advanced feature engineering, ensemble learning methods, and data balancing strategies.

In this study, an automated arrhythmia classification system is developed using the MIT-BIH Arrhythmia Database [3]. The overall framework is designed to systematically process ECG signals and evaluate the effectiveness of different machine learning models under balanced and imbalanced data conditions. The proposed framework consists of the following stages:

- ECG Preprocessing and Feature Engineering:Noise removal, R-peak detection, heartbeat segmentation, and extraction of temporal and morphological ECG features.
- Data Preparation and Classification:Feature normalization, data balancing using SMOTE, and classification using ML models such as Logistic Regression, Random Forest, LightGBM, and XGBoost.
- Model Evaluation: Performance evaluation using accuracy, precision, recall, F1-score, and confusion matrix analysis.

The objective of this work is to compare different machine learning algorithms, analyze the influence of SMOTE on classification performance, and identify the most effective approach for arrhythmia detection. The insights gained from this comparative analysis contribute toward the development of reliable ECG-based decision-support systems that may assist in the early identification of cardiac abnormalities.

III. Background

A. *ECG Signals and Their Components*

An ECG is a non-invasive recording of the heart's electrical activity and is widely used for detecting cardiac disorders. A standard ECG waveform includes the P-wave, QRS complex, and T-wave, each representing different phases of cardiac depolarization and repolarization. The RR interval, defined as the time between consecutive R-peaks, is frequently used for rhythm analysis and arrhythmia detection [3]. Variations in these waveform components often indicate abnormalities in the heart's conduction system and form the basis for automated heartbeat classification.

B. *Arrhythmia Categories*

Arrhythmias occur when the heart's electrical impulses deviate from the normal conduction pathway, resulting in abnormal or irregular rhythms. The AAMI standard categorizes ECG beats into five groups:

- **N**: Normal beats
- **S**: Supraventricular ectopic beats
- **V**: Ventricular ectopic beats
- **F**: Fusion beats
- **Q**: Unclassifiable or noisy beats

These categories are widely used in studies involving the MIT-BIH Arrhythmia Database, enabling standardized evaluation of automated classification approaches [10].

C. Class Imbalance in ECG Data

One of the key challenges in arrhythmia detection is the strong class imbalance present in ECG datasets. In the MIT-BIH dataset, normal beats occur far more frequently than abnormal beats, causing machine learning models to become biased toward the majority class. This imbalance leads to poor sensitivity and low recall for minority arrhythmias, making reliable detection difficult in practical applications [9]. Addressing this issue is essential to ensure balanced learning and improved classification performance.

D. SMOTE

The SMOTE is one of the most commonly used methods for addressing class imbalance. SMOTE generates synthetic minority samples by interpolating between existing minority instances in the feature space, thereby improving the representation of rare classes [9]. Studies applying SMOTE to arrhythmia datasets have reported higher recall and F1-scores for minority beats, demonstrating its effectiveness in improving classifier performance on imbalanced ECG data [4].

IV. Methodology

A. Dataset Description

This study combines multiple publicly available ECG datasets to create a diverse and representative set for arrhythmia classification. Data are obtained from the MIT-BIH Arrhythmia, MIT-BIH Supraventricular Arrhythmia, and St. Petersburg INCART databases. These sources provide beat-level annotations, varied sampling rates, and multi-lead recordings, improving class balance and morphological diversity, which supports more robust model training [3], [18].

B. Preprocessing

The MIT-BIH database provides ECG signals that have already undergone basic processing, including noise filtering, R-peak identification, and beat segmentation. To further enhance the quality and consistency of the data, we applied additional preprocessing steps before analysis. The procedures used are described below:

- Feature extraction: Compute temporal and morphological features including RR interval, QRS duration, PR and QT intervals, P/T amplitude measures, energy, and simple shape descriptors.
- Normalization: Standardize features using Z-score normalization to place all features on a comparable scale prior to training.

C. Data Balancing

The combined datasets exhibit significant class imbalance, with Normal (N) beats outnumbering arrhythmic categories (S, V, F, Q). Training directly on imbalanced data biases classifiers toward the majority class. To address this, experiments are carried out in two settings:

- **Without SMOTE:** Use the original imbalanced training data.
- **With SMOTE:** SMOTE on the training split to generate synthetic minority-class samples and balance the class distribution [4], [9].

Fig. 2 shows the application of the SMOTE technique to handle class imbalance in ECG data. The original dataset contains many majority-class samples and very few minority-class beats, which can bias model learning. SMOTE addresses this by generating synthetic minority samples through interpolation between nearest neighbors, improving class representation without simple duplication. The resulting balanced dataset helps classifiers learn more reliable decision boundaries.

To ensure fair evaluation, SMOTE is applied only to the training set, while validation and test data remain unchanged.

D. Machine Learning Models

We evaluate four supervised learning algorithms chosen for their complementary strengths:

- **Logistic Regression (LR):** Linear baseline model for interpretable comparisons.
- **Random Forest (RF):** Ensemble of decision trees that handles heterogeneous features and provides feature importance estimates [16].
- **LightGBM:** A fast, memory-efficient gradient boosting framework optimized for large feature sets [7].
- **XGBoost:** A highly efficient, regularized gradient boosting implementation with strong performance on structured data [8].

E. Training and Validation

Model development follows these steps:

1) **Train/test split:** Partition the data into disjoint training and test sets, preserving class proportions where possible.
2) **Cross-validation:** Use stratified K-fold cross-validation on the training set to estimate generalization performance and guide hyperparameter tuning.
3) **Hyperparameter tuning:** Perform grid search (or randomized search) over model-specific hyperparameters such as learning rate, tree depth, number of estimators, and regularization terms.

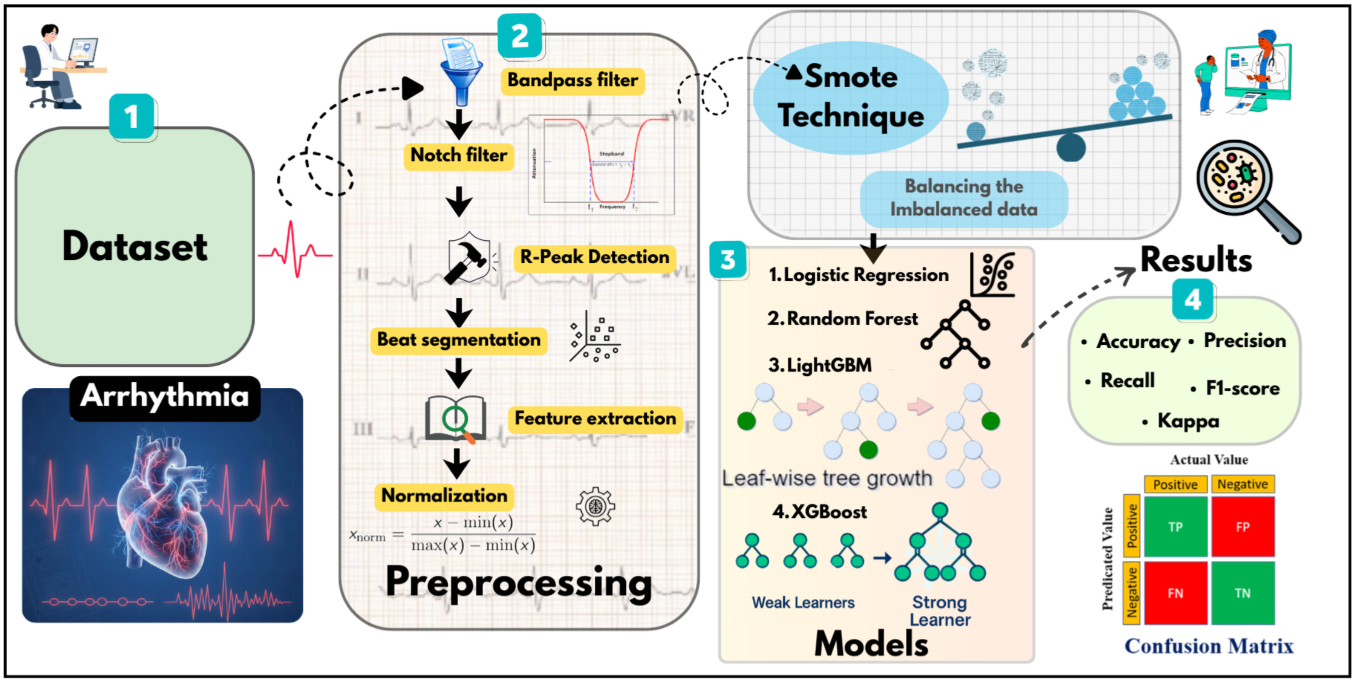

Figure 1: Overall Block Diagram of the Proposed System

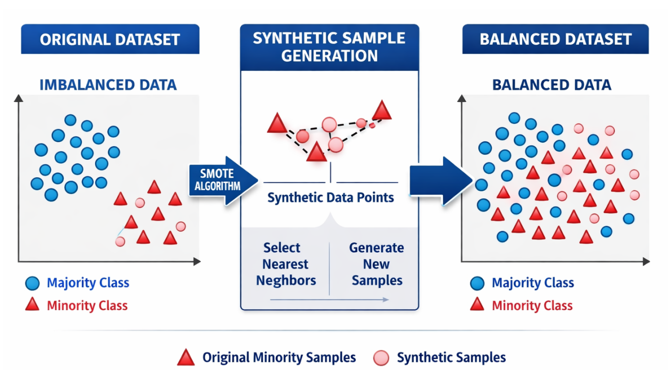

Figure 2: Illustration of the SMOTE Technique

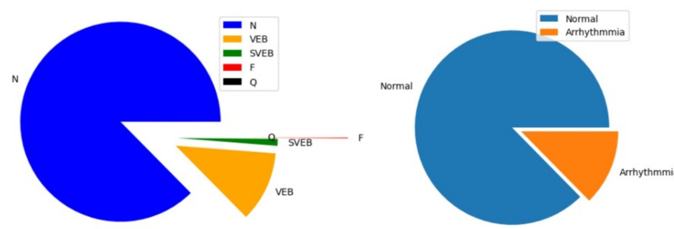

Figure 3: Pie-Chart Before Smote

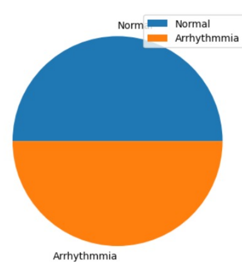

Figure 4: Pie-Chart After Smote

4) **Evaluation metrics:** Report Accuracy, Precision, Recall, F1-score (per-class and macro), AUC-ROC, and Cohen's Kappa. Confusion matrices are examined to understand per-class errors.

V. Experimental Results

This section presents experimental findings for the arrhythmia classification pipeline. We report class distributions before and after SMOTE, model-wise performance metrics, confusion matrices, and a discussion of the observed trends.

A. Class distribution before SMOTE

Before oversampling, the combined dataset shows a severe imbalance with Normal (N) beats dominating the distribution. This imbalance negatively affects minority-class detection (S, V, F, Q). Figure 3 illustrates the class distribution of the ECG dataset before applying SMOTE. The pie chart clearly shows a strong dominance of Normal beats, while arrhythmic samples constitute a much smaller portion of the dataset. This imbalance can bias classification models toward the majority class, leading to reduced detection performance for arrhythmia cases.

B. Class distribution after SMOTE

After applying SMOTE to the *training* set, class proportions are balanced for training purposes while the test set remains untouched to preserve a realistic evaluation.

C. Model performance (Accuracy, Precision, Recall, F1)

Table I: Evaluation Metrics (Accuracy, Recall and F1) for the MIT-BIH Dataset

Model	Setting	Accuracy	Recall	F1
Logistic Regression	No SMOTE	0.88	0.60	0.62
Logistic Regression	With SMOTE	0.9674	0.9680	0.9783
Random Forest	No SMOTE	0.92	0.68	0.70
Random Forest	With SMOTE	0.9913	0.9976	0.9977
XGBoost	No SMOTE	0.94	0.74	0.76
XGBoost	With SMOTE	0.9963	0.9966	0.9966
LightGBM	No SMOTE	0.93	0.70	0.73
LightGBM	**With SMOTE**	**0.9984**	**0.9987**	**0.9985**

Table I, presents a comparative performance analysis of the evaluated machine learning models with and without SMOTE. Across all models, the inclusion of SMOTE leads to noticeable improvements in recall and F1-score, indicating enhanced detection of minority arrhythmia classes. LGBM with SMOTE achieves the highest overall performance, attaining an accuracy of 99.84% along with superior recall and F1-score, demonstrating its effectiveness in handling imbalanced ECG data. These results highlight the combined advantage of ensemble learning and data balancing in improving arrhythmia classification performance.

Figure 5, Performance comparison of machine learning models trained with SMOTE-balanced data. Ensemble methods outperform Logistic Regression, with LightGBM achieving the best overall accuracy, recall, F1-score, and kappa, indicating improved minority-class detection.

Figure 6, Performance of classifiers without SMOTE balancing. Although ensemble models still perform better than Logistic Regression, reduced recall and kappa values highlight the negative impact of class imbalance on classification results.

D. Confusion matrix

Figure 7 shows the confusion matrices of the classifiers trained using SMOTE-balanced data. Compared to the imbalanced case, a clear reduction in false negatives for arrhythmic beats is observed, indicating improved sensitivity. Ensemble models, particularly LGBM and XGBoost, demonstrate better discrimination between normal and arrhythmic classes, while Logistic Regression still exhibits relatively higher misclassification. These results confirm the effectiveness of SMOTE in enhancing minority-class detection.

E. Comparative Analysis

The experimental results show the following key trends:

- **Effect of SMOTE:** Training with SMOTE increases recall and F1-scores for minority classes (S, V, F, Q), reducing clinically significant false negatives. However, SMOTE may slightly affect overall accuracy depending on test distribution.
- **Model comparison:** Boosting methods (LightGBM, XGBoost) generally outperform the simpler baselines (Logistic Regression, Random Forest) on both balanced and imbalanced training. LightGBM frequently achieves the best balance between precision and recall due to its regularized boosting mechanism.
- **Per-class behavior:** Without SMOTE, many minority-class instances are misclassified as Normal, as evidenced by the confusion matrix in Fig. **??**. With SMOTE, the confusion matrix shows improved true-positive rates for minority classes.
- **Practical implications:** For clinical monitoring, higher recall on arrhythmic classes is crucial. Although overall accuracy is useful, recall and F1 for minority classes are more relevant for patient safety.

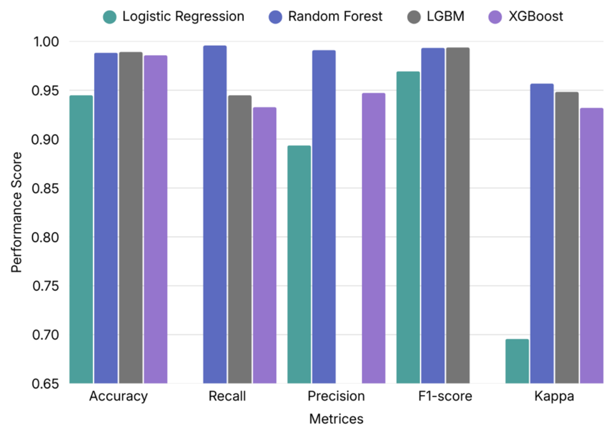

Figure 5: Performance Matrix Evaluation WITH SMOTE

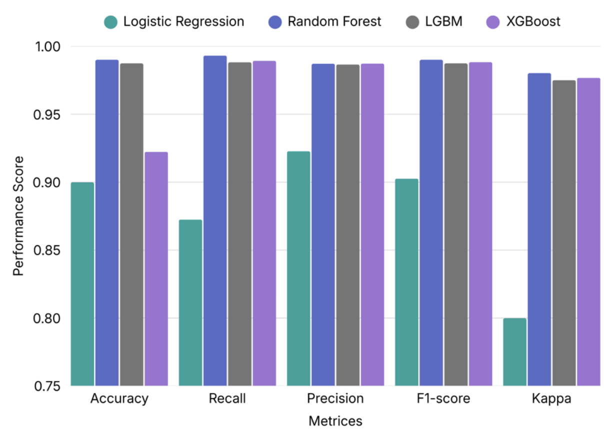

Figure 6: Performance Matrix Evaluation Without Smote

Table II presents a comparative summary of existing ECG-based arrhythmia classification approaches using the MIT-BIH database. Prior studies have demonstrated high classification performance using deep learning, ensemble, and optimized machine learning models, with reported accuracies typically ranging between 90% and 99%. The final

Paper / Author	Model Used	Dataset	Reported Performance	Ref
Acharya et al.	CNN-based heartbeat	MIT-BIH	Accuracy ≈ 94–95%	[19]
de Chazal et al.	Feature-based ML (SVM/Layout classifiers)	MIT-BIH	Accuracy ≈ 92%	[10]
Kiranyaz et al.	1D CNN (patient-specific)	MIT-BIH	>95% accuracy	[11]
Mehta & Lingayat	SVM/ANN	MIT-BIH	Accuracy 90–93%	[12]
Meliou et al.	SMOTE variants	MIT-BIH	F1-score +12–15%	[20]
Proposed	**LGBM**	**MIT-BIH**	**Accuracy 99.84%**	

Table II: Comparison of existing MIT-BIH ECG classification research.

Figure 7: Confusion Matrix with smote

row of the table corresponds to the proposed approach, which achieves an accuracy of 99.84% using LGBM combined with SMOTE. While several deep learning models report marginally higher accuracy, the proposed method offers competitive performance using a simpler machine learning framework, highlighting its effectiveness and suitability for practical ECG analysis with reduced computational complexity. Overall, these experiments validate that (1) SMOTE is an effective strategy to mitigate class imbalance in ECG datasets, and (2) ensemble boosting models—particularly LGBM—provide the most reliable performance given the features used in this study.

VI. Conclusion and Future Work

This work proposes an automated ECG arrhythmia classification system built using machine learning techniques and multiple PhysioNet datasets. The framework performs signal preprocessing, R-peak detection, beat segmentation, feature extraction, and class balancing using SMOTE before evaluating different classifiers. Handling dataset imbalance proved crucial for accurately identifying minority arrhythmia types such as supraventricular and ventricular ectopic beats. Among all models tested, LightGBM delivered the best performance in terms of accuracy, recall, and F1-score, achieving an overall accuracy of 99.84

However, the approach relies on handcrafted features and may be sensitive to noise, signal variations, and differences across datasets, which can affect generalization and real-time use. Future improvements will focus on deep learning models like CNNs and RNNs to learn features directly from raw ECG signals, along with multi-lead fusion, domain adaptation, and patient-specific optimization. Implementing the system in wearable or continuous monitoring devices is also planned to enable reliable real-time arrhythmia detection.

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